

# SmartFind

## Lost & Found Management System

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### User Research Report

#### TEAM MEMBERS

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#### COURSE

Human-Computer Interaction (HCI)

#### GROUP

Group —

## 2. Executive Summary

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SmartFind is a web-based Lost and Found Management System designed to address the inefficiencies of traditional lost-and-found processes within educational institutions. Students who lose personal belongings often rely on social media posts, messaging groups, or word-of-mouth communication, resulting in delays, confusion, and low recovery rates. Likewise, individuals who find lost items frequently lack a structured method for returning them to their rightful owners.

This project aims to centralize lost-item reporting, found-item submissions, ownership claims, and administrative oversight within a single digital platform. By streamlining communication between students, finders, and administrators, SmartFind reduces frustration, improves transparency, and increases the likelihood of successful item recovery.

This report documents the team's Human-Computer Interaction (HCI) research journey conducted between Week 2 and Week 6. Activities included problem identification, contextual inquiry, user interviews, observation of current practices, data synthesis, empathy mapping, persona development, and the creation of an initial low-fidelity workflow prototype. Particular emphasis was placed on understanding the experiences of students attempting to report and recover lost items. The findings generated actionable design requirements that will guide future prototyping and usability testing efforts.

### 3. Project Introduction & Background

SmartFind is an integrated Lost and Found Management System developed to simplify the process of reporting, searching, and recovering lost belongings within a university environment. The ecosystem operates through a connected five-role workflow:

|                                                   |                                                                                                                                                     |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Student (Lost Item Reporting)</b>              | Students who lose personal belongings create a lost-item report by providing details such as item name, category, location, description, and photo. |
| <b>Finder (Found Item Submission)</b>             | Students or staff members who discover lost items submit found-item reports that become searchable within the system.                               |
| <b>Owner (Search and Claim)</b>                   | The owner searches the platform for matching items and submits a claim request if a potential match is found.                                       |
| <b>Administrator (Verification and Oversight)</b> | Administrators review reports, verify claims, manage disputes, and update item statuses.                                                            |
| <b>System (Centralized Coordination)</b>          | The platform coordinates information flow between all users and maintains accurate records throughout the recovery process.                         |

This report focuses specifically on optimizing the user experience of students reporting lost items and searching for potential matches, as this represents the most common and critical interaction within the system.

## 4. Methodology & The HCI Journey (Week 2 – Week 6)

### Week 2 Group Formation & Problem Identification

The project team was formed with the objective of identifying a meaningful real-world problem that could benefit from improved user experience design. Through brainstorming and discussion, the team recognized that many students frequently lose personal belongings on campus and struggle to recover them due to fragmented communication channels.

Current solutions often depend on Facebook groups, messaging applications, and informal announcements, creating inefficiencies and reducing visibility. This motivated the team to explore a centralized digital solution.

### Week 3 User Research Planning & Contextual Inquiry

The team planned a research strategy focused on understanding how students currently report, search for, and recover lost items.

#### Research Objectives:

- Understanding current reporting methods
- Identifying pain points during recovery
- Examining communication barriers
- Discovering opportunities for system improvement

Contextual inquiry involved observing how students use social media groups and campus communication channels when reporting lost belongings.

**Week 4      User Interviews & Observations**

Semi-structured interviews were conducted with students who had either lost or found personal belongings. Several recurring themes emerged:

**Lost Item Owners:**

- Difficulty reaching a large audience quickly
- Uncertainty about whether an item had been found
- Excessive time spent searching multiple platforms

**Item Finders:**

- No centralized place to report found items
- Limited ability to contact the owner
- Lack of confidence that the item would be returned

**General Observations:**

- Information was often incomplete
- Posts became difficult to locate over time
- Communication relied heavily on manual effort

**Week 5      Data Synthesis — Affinity Diagramming & Empathy Mapping**

The collected qualitative data was organized using affinity diagramming techniques. Major themes included:

**Communication Challenges:**

- Users struggled to share and discover information efficiently.

**Trust and Verification:**

- Users wanted confidence that items would be returned to legitimate owners.

**Search Efficiency:**

- Students desired faster methods for locating matching items.

**Process Visibility:**

- Users wanted updates regarding item status and claim progress.

Empathy mapping revealed that students frequently experienced frustration, anxiety, uncertainty, and helplessness after losing valuable belongings.

| Week 6 | User Personas & Initial Low-Fidelity Flow                                                                                                                                                         |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | <p>The research findings were synthesized into a primary user persona representing a typical student. An initial low-fidelity workflow was developed to address the most critical user needs.</p> |
|        | <p><b>Low-Fidelity Workflow:</b></p>                                                                                                                                                              |
|        | <ul style="list-style-type: none"><li>• Screen 1 — Lost Item Dashboard: search bar, recent found items, Report Lost Item button</li></ul>                                                         |
|        | <ul style="list-style-type: none"><li>• Screen 2 — Lost Item Report Form: item details, location, description, photo upload, submit button</li></ul>                                              |
|        | <p>This workflow was intentionally simplified to minimize user effort while maximizing reporting efficiency.</p>                                                                                  |

## 5. User Persona & Workflow Specification

### User Persona

| Attribute        | Details                                                                                                   |
|------------------|-----------------------------------------------------------------------------------------------------------|
| Name             | Sokha                                                                                                     |
| Age              | 19                                                                                                        |
| Occupation       | University Student                                                                                        |
| Goals            | Recover lost belongings quickly<br>Report lost items efficiently<br>Receive updates regarding item status |
| Frustrations     | Searching across multiple platforms<br>Lack of response from finders<br>Unclear recovery process          |
| Technology Usage | Daily smartphone user<br>Active on social media<br>Comfortable using web applications                     |

### Primary User Story

*"As a student, I want to report a lost item and search for matching found items so that I can recover my belongings quickly and efficiently."*

### Workflow Specification

|                     |                                                                                                                                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Starting Point      | The student opens the SmartFind homepage and selects "Report Lost Item."                                                                                                  |
| Successful Endpoint | The student submits a lost-item report successfully and receives a confirmation message indicating that the report has been recorded and is searchable within the system. |

## 6. Key Insights & Design Implications

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The research generated five primary design requirements that will guide the development of SmartFind's user interface and interaction model:

|   |                                                                                                                                                   |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Minimize Data Entry Effort</b><br>Forms should require only essential information to encourage participation and reduce abandonment.           |
| 2 | <b>Improve Search Visibility</b><br>Search and filtering tools should help users locate relevant items quickly.                                   |
| 3 | <b>Provide Clear Status Indicators</b><br>Users should be able to distinguish between Lost, Found, Claimed, and Returned items at a glance.       |
| 4 | <b>Support Trust and Verification</b><br>The claim process should provide sufficient evidence requirements to reduce fraudulent ownership claims. |
| 5 | <b>Maintain Simple Navigation</b><br>Core actions such as reporting, searching, and claiming should be accessible within a few clicks.            |



## 7. Conclusion & Next Steps

This report documented the HCI research journey undertaken from Week 2 through Week 6 for the SmartFind Lost and Found Management System. Through problem identification, contextual inquiry, interviews, observations, affinity diagramming, empathy mapping, and persona development, the team gained a deeper understanding of the challenges students face when attempting to recover lost belongings.

The findings highlighted significant opportunities to improve communication, search efficiency, transparency, and trust within the lost-and-found process. These insights informed the creation of a user-centered workflow designed to streamline item reporting and recovery.

The next phase of the project involves transforming these research findings into interactive prototypes, followed by usability testing sessions to evaluate effectiveness, identify usability issues, and refine the user experience before further development.

| Phase   | Activity                          | Goal                                                       |
|---------|-----------------------------------|------------------------------------------------------------|
| Phase 1 | Interactive Prototype Development | Translate low-fidelity flows into clickable prototypes     |
| Phase 2 | Usability Testing                 | Evaluate prototype effectiveness with target users         |
| Phase 3 | Iteration & Refinement            | Address usability issues and improve interaction design    |
| Phase 4 | Final Development                 | Build the full SmartFind system based on validated designs |